

Series completion

COMPLETE LEARNING GUIDE

What You'll Learn

Series completion questions — number series, letter series, alphanumeric series, and figure series — test pattern recognition by presenting a sequence with one clear underlying rule and asking you to find the next term, an odd term, or a deliberately wrong term planted in the sequence. The reliable method is always the same: compute the difference (or ratio) between consecutive terms first, and if that doesn't reveal a clean pattern, check for an alternating two-step rule or a polynomial (n^2 -based) pattern before assuming the series is unsolvable. This sub-module works through number series with constant and increasing differences, letter series with alphabet-position jumps, alphanumeric combinations, figure series described conceptually, odd-one-out detection, alternating two-step series, and wrong-number-in-the-series questions — each solved completely with the underlying rule stated explicitly.

1. Number Series: Difference and Ratio Patterns

Start by computing the difference between consecutive terms; if that sequence of differences is itself constant or follows a simple pattern (like consecutive odd numbers), the rule is revealed. If differences don't work, try ratios (each term $\times/\div$ a constant or increasing factor).

- 2, 5, 10, 17, 26, ? — differences: 3, 5, 7, 9 (consecutive odd numbers, increasing by 2 each time). Next difference = 11. Next term = $26 + 11 = 37$.

2. Letter Series

Letter series follow the same difference-based logic, but the 'difference' is measured in alphabet positions (A=1, B=2, ... Z=26) rather than numeric value.

- A, C, F, J, O, ? — position gaps: +2, +3, +4, +5, +6. O is the 15th letter; +6 gives the 21st letter, U. So the next term is U.

3. Alphanumeric Series

These combine a letter pattern and a number pattern running in parallel within the same sequence — solve each sub-pattern (letters, numbers) independently, since they usually follow separate, simpler rules.

- A1, C3, E5, G7, ? — letters skip one each time (+2 position: A!C!E!G!I) and numbers increase by 2 (odd numbers: 1,3,5,7!9). Next term: I9.

4. Figure Series

Figure series apply the same 'find the transformation rule' logic to shapes instead of numbers or letters — common transformations include rotation by a fixed angle, addition/removal of an element, or a shape cycling through a fixed set of positions — and the correct next figure must continue that exact same transformation, not just look visually similar to the others.

5. Odd One Out in a Series

Odd-one-out questions give several terms where all but one follow a clear rule — first establish the rule from the majority of terms, then check each term against it to isolate the outlier.

- 2, 4, 8, 16, 30, 64 — the rule is powers of 2 (2,4,8,16,32,64). Every term fits except 30, which should be 32. 30 is the odd one out.

6. Alternating (Two-Step) Series

Some series apply two different operations alternately (e.g., $\times 2$ then -1 , repeating) rather than one constant operation — if a single consistent difference or ratio doesn't fit, test whether alternating terms follow two separate, interleaved rules.

- 3, 6, 5, 10, 9, 18, 17, ? — pattern alternates $\times 2$ and -1 : $3 \times 2 = 6$, $6 - 1 = 5$, $5 \times 2 = 10$, $10 - 1 = 9$, $9 \times 2 = 18$, $18 - 1 = 17$, $17 \times 2 = 34$. Next term = 34.

7. Finding the Wrong Number in a Series

These questions present a full series with one term deliberately altered — find the underlying rule (often n^2 , n^2+1 , or similar) using the terms that clearly fit, then compute what the flawed term SHOULD be to identify and correct it.

- 5, 10, 17, 26, 37, 50, 64 — the rule is n^2+1 for $n=2,3,4,5,6,7,8$: 5,10,17,26,37,50,65. The given series has 64 instead of 65 — 64 is the wrong number.

Common Pitfalls

- Only checking for a constant difference and giving up if it isn't found, instead of also checking ratios, alternating patterns, or polynomial (n^2) rules.
- In letter series, forgetting to convert letters to their numeric alphabet position before computing gaps.
- In alphanumeric series, trying to find one combined rule instead of solving the letter and number sub-patterns separately.
- In odd-one-out questions, establishing the rule from a minority of terms (including the actual outlier) instead of the clear majority.

- Missing an alternating two-step pattern by only testing a single constant operation between all consecutive terms.
- In wrong-number questions, assuming the LAST term is always the error instead of testing the rule against every term.

Key Takeaways

- Always start with differences between consecutive terms; if that fails, test ratios, then alternating two-step rules, then n^2 -based patterns.
- Convert letters to alphabet positions (A=1...Z=26) before analyzing letter-series gaps.
- Alphanumeric series usually hide two independent, simpler sub-patterns (letters and numbers) rather than one combined rule.
- Establish a series rule from its clear majority of terms before testing which single term breaks it.
- Alternating series apply two different operations in a repeating two-step cycle — test this whenever a single operation doesn't fit.
- Figure series require identifying the exact transformation rule (rotation angle, added element), not just visual similarity to prior figures.

Learn More

IndiaBix — Non-Verbal Reasoning: Series

<https://www.indiabix.com/non-verbal-reasoning/series/>

GeeksforGeeks — Non-Verbal Reasoning Questions and Answers

<https://www.geeksforgeeks.org/non-verbal-reasoning-questions-and-answers/>